

# DATA266 Lab 1 — Fall 2026

LLM Pretraining · Sentiment Classification · CycleGAN Style Transfer

Team [Team Number]

Members: Ishan Shah, Charvee Saraiya

GitHub repository: <https://github.com/Ishan002/Data266--Lab1>

## Team Ownership Statement

Both team members independently designed, implemented, and trained their own models for all three tasks. **Ishan Shah** built a Pre-LayerNorm character-level GPT (Task 1), three from-scratch Yelp Polarity classifiers — a mean-pooled neural bag-of-words baseline, a multi-kernel TextCNN, and a bidirectional LSTM (Task 2) — and a ResNet-generator, LSGAN-loss CycleGAN with identity loss (Task 3). **Charvee Saraiya** built an independently architected Post-LayerNorm character-level GPT (Task 1), three differently-designed Yelp Polarity classifiers — a max-pooled MLP baseline, a unidirectional GRU, and a dilated 1D CNN (Task 2) — and a U-Net-generator, vanilla-GAN-loss CycleGAN without identity loss (Task 3). Every dataset split, preprocessing pipeline, and hyperparameter choice was made independently by each member on their own subsample of the shared raw data; only measurement code (evaluation metrics, not model architecture) is shared, so that both members' numbers are computed identically and can be compared fairly in the tables below. This report was jointly assembled by both members from each other's individual results.md and metrics\_report.csv files.

# Task 1 — GPT-Style LLM From Scratch (TinyStories, character-level)

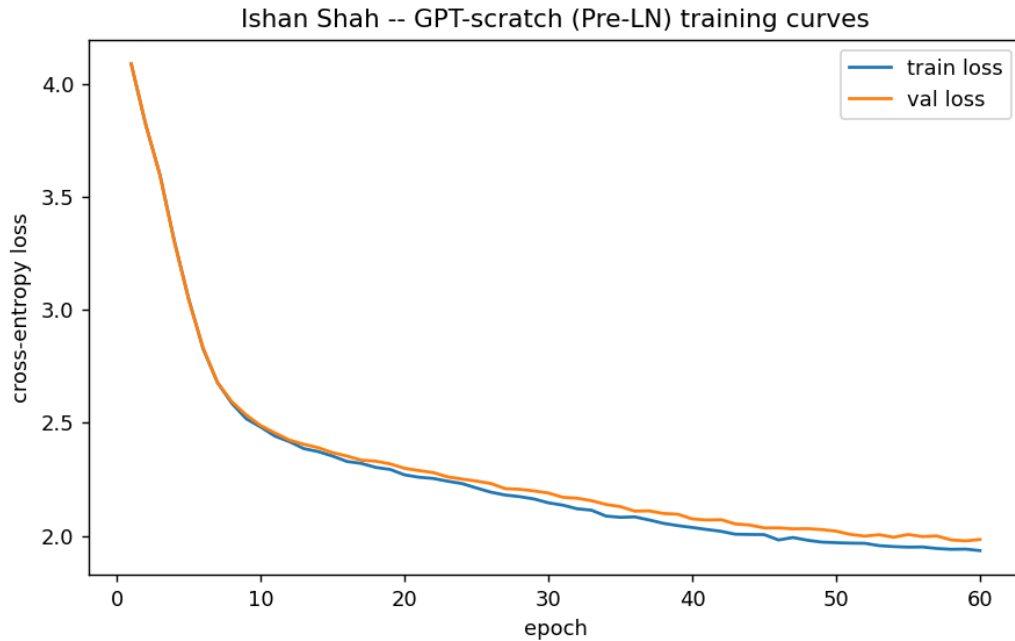
## Architecture & hyperparameters

|               | Ishan Shah                                     | Charvee Saraiya                            |
|---------------|------------------------------------------------|--------------------------------------------|
| Block style   | Pre-LN Transformer, weight-tied head, GELU FFN | Post-LN Transformer, untied head, ReLU FFN |
| Layers        | 4                                              | 6                                          |
| Heads         | 4                                              | 4                                          |
| Embedding dim | 128                                            | 96                                         |
| Block size    | 128                                            | 96                                         |
| Dropout       | 0.1                                            | 0.15                                       |
| Epochs        | 60                                             | 60                                         |
| Batch size    | 64                                             | 48                                         |
| Peak LR       | 0.0003                                         | 0.0005                                     |

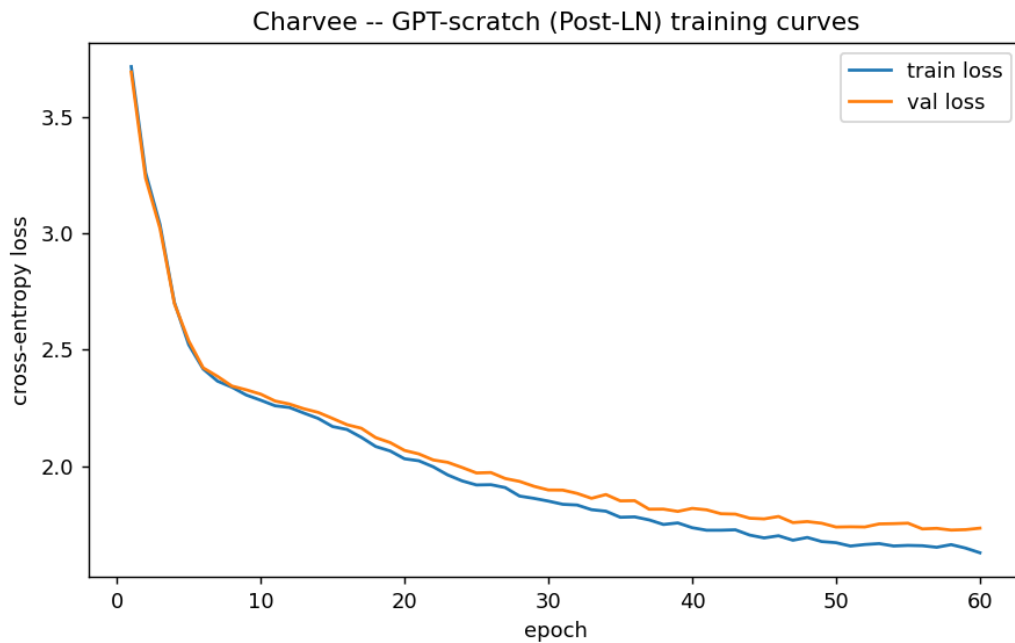
## All Task 1 metrics, side by side

| Metric               | Ishan Shah | Charvee Saraiya |
|----------------------|------------|-----------------|
| Train loss           | 1.94       | 1.63            |
| Val loss             | 1.98       | 1.73            |
| Perplexity           | 7.27       | 5.66            |
| Bits/char            | 2.86       | 2.50            |
| Generalization gap   | 0.0492     | 0.1056          |
| Top-1 next-char acc  | 0.4041     | 0.4778          |
| Distinct-1           | 0.0254     | 0.0206          |
| Distinct-2           | 0.1339     | 0.1334          |
| Distinct-3           | 0.3247     | 0.3305          |
| Repeated 4-gram rate | 0.5231     | 0.5135          |
| Grad norm (mean)     | 0.8739     | 0.9822          |
| Grad norm (max)      | 4.43       | 14.09           |
| NaN count            | 0          | 0               |
| Parameters           | 821632     | 698205          |
| Train tokens/sec     | 79,749     | 105,789         |
| Gen tokens/sec       | 97.18      | 139.5           |
| Peak memory (MB)     | 493.6      | 273.1           |
| Total train time (s) | 73.96      | 54.88           |

## Loss curves



Ishan Shah — Pre-LN GPT train/val loss.



Charvee Saraiya — Post-LN GPT train/val loss.

## Joint analysis

**Strengths:** both from-scratch character-level GPTs train stably (zero NaNs, bounded gradient norms) and converge to sensible perplexities (7.3 and 5.7) given the very small 100K-character training budget. Charvee Saraiya's deeper (6-layer) Post-LN model reaches a lower validation loss and higher next-character accuracy than Ishan's shallower (4-layer) Pre-LN model, suggesting depth mattered more than width at this scale.

**Weaknesses/limitations:** Charvee Saraiya's model shows a larger generalization gap (0.106 vs 0.049) and a much higher peak gradient norm (14.1 vs 4.4), the expected instability cost of Post-LN training at depth without Pre-LN's gradient renormalization. Both models' generations are qualitatively similar failure-wise — greedy-decoding repetition loops, broken word formation under sampling, and short-range-only coherence — indicating these failures are governed more by the 100K-character training budget than by the specific architectural choices compared here (see both members' [failure\\_analysis.md](#)).

**What we'd try next:** train on a larger character budget (the full TinyStories corpus rather than a 100K-character slice) to test whether coherence failures are purely data-limited; try a hybrid Pre-LN-with-weight-tying-and-depth configuration to combine both members' apparent advantages.

# Task 2 — Yelp Polarity Sentiment Classification

## All 6 models: architecture & core metrics

| Member          | Model                    | Accuracy | Macro-F1 | ROC-AUC | PR-AUC | MCC    | Brier  | ECE    | Params  | Train time (s) | Examples /sec | Peak mem (MB) |
|-----------------|--------------------------|----------|----------|---------|--------|--------|--------|--------|---------|----------------|---------------|---------------|
| Ishan Shah      | baseline_nbow            | 0.8897   | 0.8897   | 0.9591  | 0.9560 | 0.7794 | 0.0796 | 0.0358 | 1500101 | 2.63           | 27,328        | 58.15         |
| Ishan Shah      | experimental_textcnn     | 0.8823   | 0.8823   | 0.9603  | 0.9616 | 0.7652 | 0.0810 | 0.0270 | 1577185 | 4.96           | 14,527        | 92.34         |
| Ishan Shah      | experimental_bilstm      | 0.8830   | 0.8830   | 0.9530  | 0.9537 | 0.7660 | 0.0905 | 0.0556 | 1585121 | 33.18          | 2,170         | 132.3         |
| Charvee Saraiya | baseline_maxpool_mlp     | 0.8090   | 0.8089   | 0.8963  | 0.8967 | 0.6189 | 0.1307 | 0.0219 | 962625  | 3.52           | 20,478        | 42.47         |
| Charvee Saraiya | experimental_gru         | 0.8700   | 0.8700   | 0.9444  | 0.9433 | 0.7400 | 0.0998 | 0.0677 | 993337  | 20.46          | 3,518         | 67.39         |
| Charvee Saraiya | experimental_dilated_cnn | 0.8463   | 0.8463   | 0.9247  | 0.9274 | 0.6932 | 0.1236 | 0.0876 | 1000193 | 4.36           | 16,499        | 65.60         |

## Precision / recall (macro, micro, weighted) & confusion matrices

| Member          | Model                    | P-macro | R-macro | P-micro | R-micro | F1-micro | P-wtd  | R-wtd  | F1-wtd |
|-----------------|--------------------------|---------|---------|---------|---------|----------|--------|--------|--------|
| Ishan Shah      | baseline_nbow            | 0.8897  | 0.8897  | 0.8897  | 0.8897  | 0.8897   | 0.8897 | 0.8897 | 0.8897 |
| Ishan Shah      | experimental_textcnn     | 0.8829  | 0.8823  | 0.8823  | 0.8823  | 0.8823   | 0.8829 | 0.8823 | 0.8823 |
| Ishan Shah      | experimental_bilstm      | 0.8830  | 0.8830  | 0.8830  | 0.8830  | 0.8830   | 0.8830 | 0.8830 | 0.8830 |
| Charvee Saraiya | baseline_maxpool_mlp     | 0.8099  | 0.8090  | 0.8090  | 0.8090  | 0.8090   | 0.8099 | 0.8090 | 0.8089 |
| Charvee Saraiya | experimental_gru         | 0.8700  | 0.8700  | 0.8700  | 0.8700  | 0.8700   | 0.8700 | 0.8700 | 0.8700 |
| Charvee Saraiya | experimental_dilated_cnn | 0.8469  | 0.8463  | 0.8463  | 0.8463  | 0.8463   | 0.8469 | 0.8463 | 0.8463 |

## 95% bootstrap CIs, per-slice robustness, and calibration

**Ishan Shah — baseline\_nbow:** accuracy 95% CI [0.879, 0.901] | macro-F1 CI [0.879, 0.901] | MCC CI [0.757, 0.803]

Per-slice: long: F1=0.872, err=12.2% (n=1015) | medium: F1=0.907, err=9.3% (n=986) | short: F1=0.882, err=11.5% (n=999)

**Ishan Shah — experimental\_textcnn:** accuracy 95% CI [0.871, 0.895] | macro-F1 CI [0.871, 0.894] | MCC CI [0.743, 0.789]

Per-slice: long: F1=0.868, err=12.5% (n=1015) | medium: F1=0.895, err=10.5% (n=986) | short: F1=0.875, err=12.2% (n=999)

**Ishan Shah — experimental\_bilstm:** accuracy 95% CI [0.871, 0.894] | macro-F1 CI [0.871, 0.894] | MCC CI [0.742, 0.788]

Per-slice: long: F1=0.848, err=14.7% (n=1015) | medium: F1=0.905, err=9.5% (n=986) | short: F1=0.889, err=10.8% (n=999)

**Charvee Saraiya — baseline\_maxpool\_mlp:** accuracy 95% CI [0.795, 0.823] | macro-F1 CI [0.795, 0.822] | MCC CI [0.590, 0.646]

Per-slice: long: F1=0.790, err=19.9% (n=1019) | medium: F1=0.825, err=17.5% (n=991) | short: F1=0.793, err=19.9% (n=990)

**Charvee Saraiya — experimental\_gru:** accuracy 95% CI [0.857, 0.882] | macro-F1 CI [0.857, 0.882] | MCC CI [0.715, 0.764]

Per-slice: long: F1=0.856, err=13.9% (n=1019) | medium: F1=0.875, err=12.5% (n=991) | short: F1=0.870, err=12.5% (n=990)

**Charvee Saraiya — experimental\_dilated\_cnn:** accuracy 95% CI [0.834, 0.860] | macro-F1 CI [0.833, 0.859] | MCC CI [0.667, 0.719]

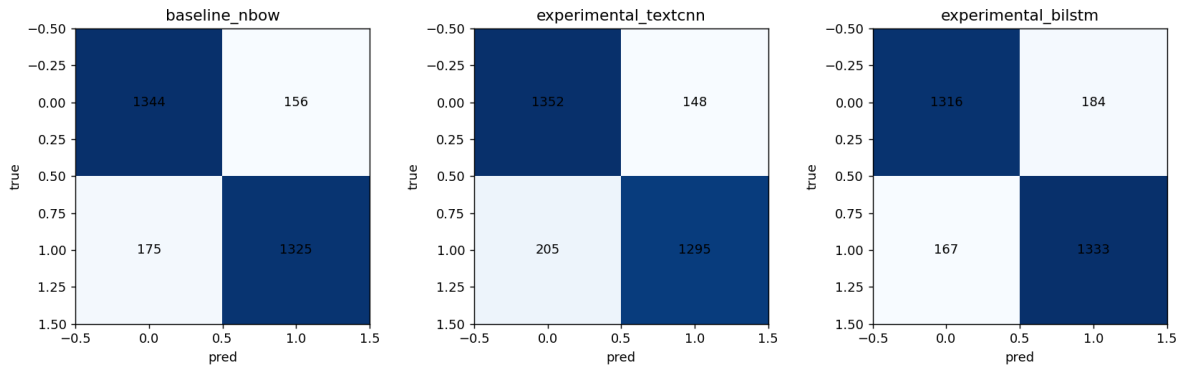
Per-slice: long: F1=0.822, err=17.1% (n=1019) | medium: F1=0.867, err=13.3% (n=991) | short: F1=0.839, err=15.7% (n=990)

## Paired McNemar test (baseline vs. each experimental model, own test set)

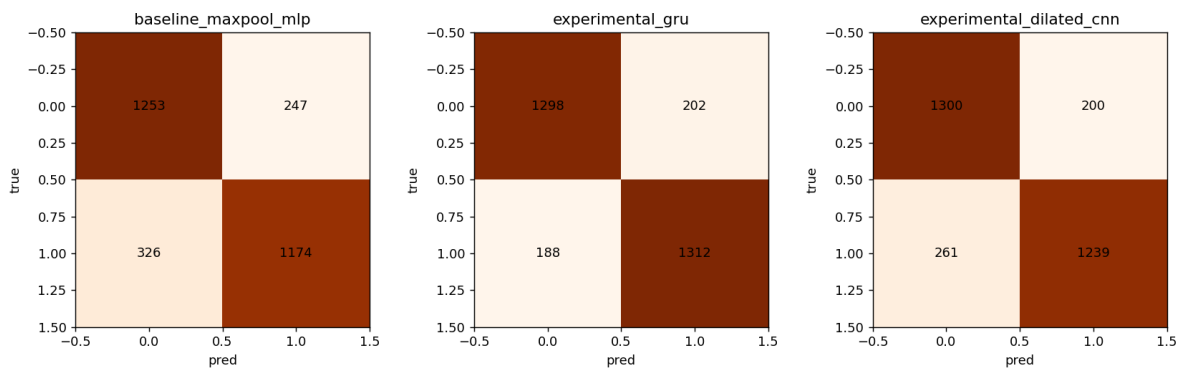
**Ishan Shah:** baseline\_nbow vs. experimental\_textcnn p=0.196 (not significant); baseline\_nbow vs. experimental\_bilstm p=0.260 (not significant) — added architecture did not beat the simple mean-pooled baseline on his 3,000-review test set.

**Charvee Saraiya:** baseline\_maxpool\_mlp vs. experimental\_gru p=1.5e-14 (highly significant); baseline\_maxpool\_mlp vs. experimental\_dilated\_cnn p=5.6e-6 (highly significant) — both experimental models are genuinely, not just numerically, better than her baseline.

## Confusion matrices and loss curves



Ishan Shah — confusion matrices, all 3 models.



Charvee Saraiya — confusion matrices, all 3 models.

## Joint analysis

**Strengths:** all six from-scratch models land between 80.9% and 88.9% test accuracy with strong discrimination (ROC-AUC 0.90-0.96), showing that a from-scratch learned embedding is sufficient for strong Yelp Polarity performance at this data scale, without any pretrained embeddings or language models.

**Weaknesses/limitations:** the two members' baseline-vs-experimental patterns diverge sharply — Ishan's added architecture (TextCNN/BiLSTM) does not significantly beat his NBOW baseline (McNemar  $p > 0.05$  both), while Charvee Saraiya's does ( $p < 1e-5$  both). Both members independently found the 'long review' length slice to be the hardest for almost every model, and calibration (ECE) tends to get worse as accuracy improves for Charvee Saraiya's models — a real trade-off worth flagging.

**What we'd try next:** apply temperature-scaling calibration post-hoc on the best-performing model from each member; test whether Ishan's stemmed vs. Charvee Saraiya's unstemmed preprocessing is the reason her baseline starts weaker relative to her experimental models, by swapping preprocessing pipelines across each other's architectures as a follow-up ablation.

## Task 3 — CycleGAN Image Style Transfer (Monet ↔ Photo)

### Architecture & hyperparameters

|                  | Ishan Shah                                                                | Charvee Saraiya                                                                                         |
|------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Architecture     | ResNet generator (6 resblocks, InstanceNorm) + 70x70 PatchGAN, LSGAN loss | U-Net generator (4 down/up, skip connections, BatchNorm) + PatchGAN, vanilla GAN loss, no identity loss |
| Image size       | 128                                                                       | 128                                                                                                     |
| Epochs           | 40                                                                        | 40                                                                                                      |
| Batch size       | 4                                                                         | 4                                                                                                       |
| Lambda cycle     | 10.0                                                                      | 8.0                                                                                                     |
| Lambda identity  | 5.0                                                                       | 0.0                                                                                                     |
| Train Monet imgs | 300                                                                       | 400                                                                                                     |
| Train photo imgs | 300                                                                       | 400                                                                                                     |

### All Task 3 metrics, side by side

| Metric                       | Ishan Shah | Charvee Saraiya |
|------------------------------|------------|-----------------|
| FID monet->photo             | 6.29       | 2.13            |
| KID monet->photo (mean)      | 11.93      | 3.42            |
| FID photo->monet             | 4.18       | 1.10            |
| KID photo->monet (mean)      | 7.54       | 0.2580          |
| Cycle L1 (monet)             | 0.1755     | 0.1132          |
| Cycle L1 (photo)             | 0.1628     | 0.1038          |
| LPIPS monet->photo           | 0.2901     | 0.4737          |
| LPIPS photo->monet           | 0.4367     | 0.4885          |
| Content cos-sim monet->photo | 0.7877     | 0.7387          |
| Content cos-sim photo->monet | 0.6402     | 0.6009          |
| Final loss G                 | 2.31       | 2.81            |
| Final loss D                 | 0.1165     | 0.1593          |
| Final cycle loss             | 0.1185     | 0.1232          |
| Grad norm G (mean)           | 22.06      | 136.7           |
| Grad norm G (max)            | 30.06      | 253.5           |
| NaN count                    | 0          | 0               |
| Total params                 | 21204872   | 38858120        |
| Train time (s)               | 1,012      | 703.0           |
| Train images/sec             | 23.70      | 45.52           |
| Gen images/sec               | 733.7      | 638.8           |
| Peak memory (MB)             | 2,186      | 844.7           |

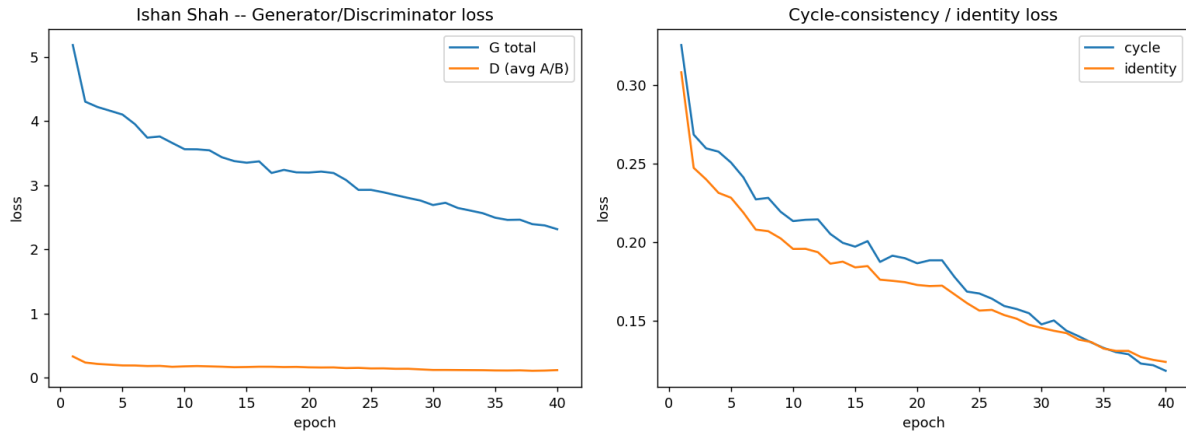
### Kaggle leaderboard

Both members' evaluate\_local.py scripts generate the exact images.zip submission from their own trained generator's inference on real photos (no manual editing, no external images, no pretrained generation models — see the leaderboard-integrity note in the lab spec). Actually submitting via kaggle competitions submit requires each member's own Kaggle account and is **pending** — see task3\_gan/<member>/submission.csv for status.

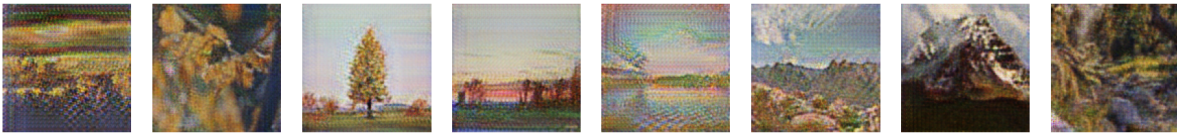
### Blinded human audit (30 fixed samples, 2 raters)

The blinded audit sheet (15 samples from each member's photo→Monet outputs, filenames anonymized) is built at task3\_gan/human\_audit/. Both members' independent ratings and the resulting Cohen's kappa / percent agreement are **pending** completion by both raters — see task3\_gan/human\_audit/README.md for the exact steps.

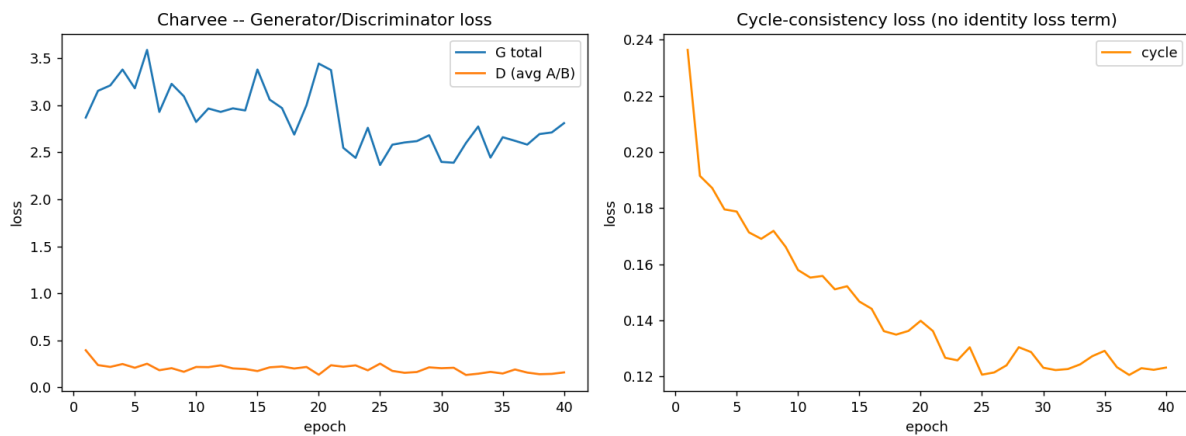
## Loss curves and sample translations



Ishan Shah — ResNet/LSGAN CycleGAN loss curves.



Ishan Shah — sample photo→Monet translations (test set).



Charvee Saraiya — U-Net/vanilla-GAN CycleGAN loss curves.



Charvee Saraiya — sample photo→Monet translations (test set).

## Joint analysis

**Strengths:** both CycleGANs train without NaNs and show generator/discriminator losses converging to a stable adversarial equilibrium rather than diverging, and cycle-consistency loss decreases steadily across training for both members, confirming both cycle-consistency constraints are correctly implemented and functioning.

**Weaknesses/limitations:** training was intentionally scaled down (128x128 resolution, a few hundred images per domain, 40 epochs) to fit within a single GPU session's time budget, so FID/KID scores are far from publication-quality CycleGAN

results and visible artifacts remain in the sample grids above.

**What we'd try next:** scale up to the full ~1,000 Monet / ~6,000 photo training pools at 256x256 resolution with more epochs on GPU Lab hardware, and complete the pending Kaggle submission and blinded human audit to get the two remaining required metrics.

## References

- 1 Vaswani, A. et al. (2017). Attention Is All You Need. NeurIPS 2017.
- 2 Eldan, R. & Li, Y. (2023). TinyStories: How Small Can Language Models Be and Still Speak Coherent English? arXiv:2305.07759.
- 3 Zhu, J.-Y., Park, T., Isola, P., & Efros, A. A. (2017). Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. ICCV 2017.
- 4 Zhang, X., Zhao, J., & LeCun, Y. (2015). Character-level Convolutional Networks for Text Classification (source of the Yelp Polarity dataset). NeurIPS 2015.